(Nyarugenge and Gako Campuses)

UNIVERSITY of RWANDA

COLLEGE OF SCIENCE & TECHNOLOGY-SCHOOL OF ICT

DEPARTMENT OF COMPUTER AND SOFTWARE ENGINEERING
END OF SEMESTER I EXAMINATION
ACADEMIC YEAR 2025-2026
MODULE: CE80363 OPERATING SYSTEMS
YEAR: 2 CSE both Nyarugenge and Gako Campuses

DATE: 16th/01/2026

TIME: 2 hours

MAXIMUM MARKS = 50

INSTRUCTIONS

1. This question paper contains **two sections: A and B.**
2. **Section A is compulsory**
3. **Choose any two questions in Section B**
4. No written materials allowed.
5. Do not forget to write your Registration Number.
6. Do not write any answers on this question paper.
7. Switch off your mobile phones
8. Start each question in a NEW page

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SECTION: A

COMPULSORY QUESTION

QUESTION: 1 **(20 Marks)**

1.1 Name and describe the different states that process can exist in at any given time ? **(2 marks)**

1.2 Explain the main difference between short-term and long term scheduler . **(2 marks)**

1.3 Distinguish different operating system user interface. **(2 Marks)**

1.4 In the context of operating system, define the following

i. Page table entry **(2 Marks)**

ii. Translation Lookaside Buffer **(2 Marks)**

iii. Belady's anomaly **(2 Marks)**

iv. Paging **(2 Marks)**

1.5 Briefly explain the following bit in page table entry: Darty bit, Referenced bit and Valid bit **(2 Marks)**

1.6 Explain the concept of context switch **(2 Marks)**

1.7 When a process is executing, it is in the running state. At any time, the OS may interrupt the process and switch to the next process that is waiting for CPU service. Describe the various reasons why a running process may be interrupted. **(2 Marks)**

SECTION B.: ANSWER ANY TWO QUESTIONS

QUESTION: 2 **(15 Marks)**

a) Using a diagram in your example, explain the 2-levels address translation architecture in memory management of a 32 bits systems among them 13 bits are used as Offset bits **(3 Marks)**

b) Explain how monitors overcome process synchronization problem **(3 Marks)**

c) Briefly discuss on CPU Scheduling algorithm in operating systems **(4 Marks)**

d) Consider the set of 5 processes whose arrival time and burst time are given below, draw Gantt chart and calculate the average waiting time and average turnaround time If the CPU scheduling policy is Preemptive Shortest Job First. **(5 Marks)**

Process ID	Arrival time	Burst time
P1	2	6
P2	5	2
P3	1	8
P4	0	3
P5	4	4

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QUESTION 3 (15 Marks)

3.1 Given a the same reference string for all algorithms as follows: 1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6 with 4 frames in physical memory for all. calculate the number of page faults and page hits using :

a) Least Recently Used page replacement algorithm **(4 Marks)**

b) Optimal Page replacement algorithm **(4 Marks)**

c) First come First Out page replacement algorithm **(4 Marks)**

3.2 Discuss about your result above and indicate the best algorithm among them **(3 Marks)**

QUESTION 4 (15 Marks)

In Operating System concepts, file systems plays a pivotal role in the exchange and storing data on both primary and secondary storage. Disks are always needs to be divided into file systems ensure proper **disk free space** management. In this regard there exists various method of file systems free space management.

a) State and explain all of them. **(4 Marks)**

(b) Suppose you are given 5 memory partitions of 220Kb, 120Kb, 520Kb, 610Kb and 300Kb in order. How would each algorithm allocate the processes of 212 Kb, 417Kb, 132Kb, and 426Kb?**(5 Marks)**

c)With a help of preemptive Priority CPU scheduling policy, for the given processes given their arrival time and bust time as follows: (the small value means the Higher priority)

Process ID	Arrival time	Priority	Burst time
P1	0	3	8
P2	1	4	2
P3	3	4	4
P4	4	5	1
P5	5	2	6
P6	6	6	5
P7	10	1	1

i) Draw Gantt chart of the processes **(2 Marks)**

ii) calculate the average turnaround time **(2 Marks)**

iii) Calculate the average of waiting time **(2 Marks)**

*********Good Luck*****

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